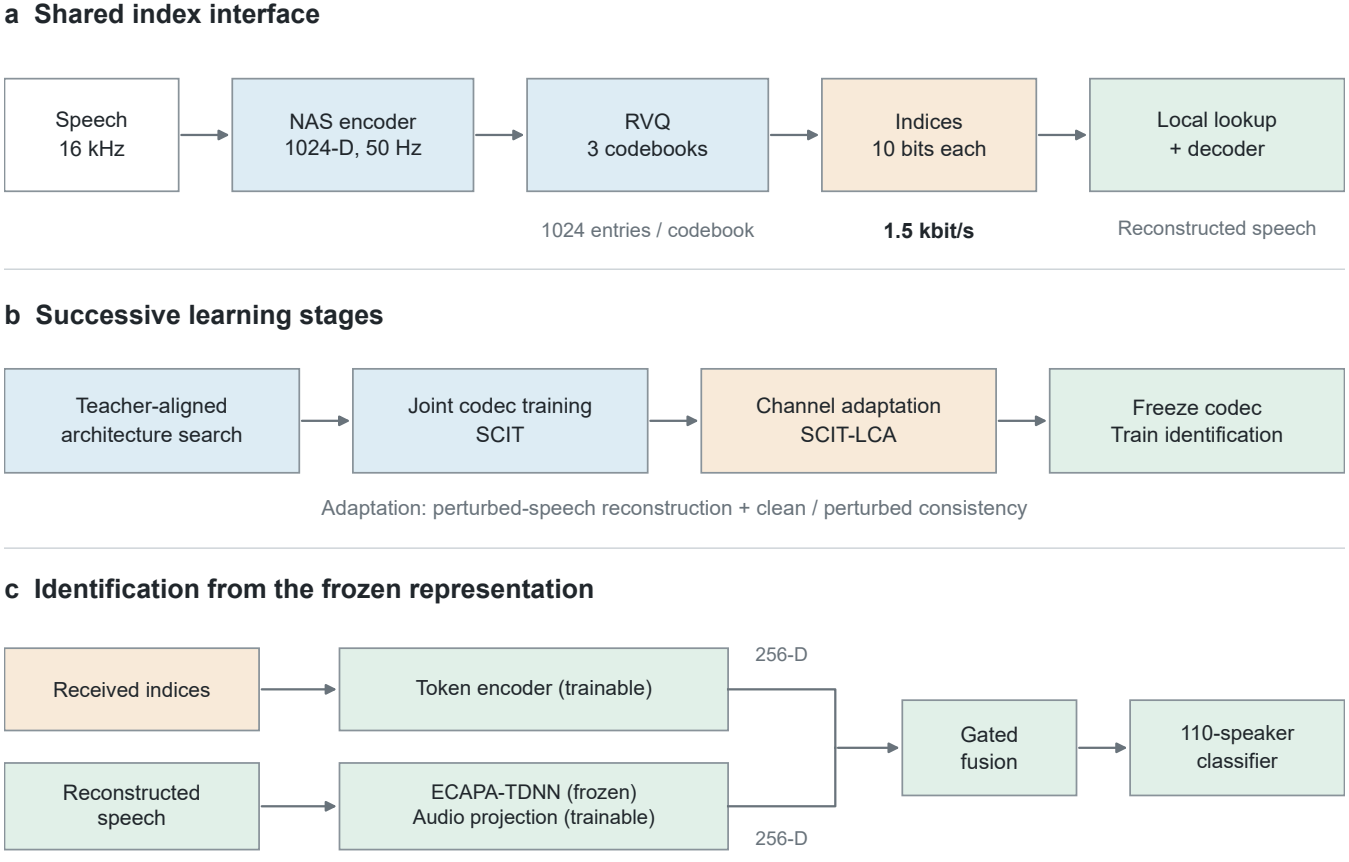
a Shared index interface
Speech
16 kHz
NAS encoder
1024-D, 50 Hz
RVQ
3 codebooks
Indices
10 bits each
Local lookup
+ decoder
1024 entries / codebook
1.5 kbit/s
Reconstructed speech
b Successive learning stages
Teacher-aligned
architecture search
Joint codec training
SCIT
Channel adaptation
SCIT-LCA
Freeze codec
Train identification
Adaptation: perturbed-speech reconstruction + clean / perturbed consistency
c Identification from the frozen representation
Received indices
Token encoder (trainable)
256-D
Reconstructed
speech
ECAPA-TDNN (frozen)
Audio projection (trainable)
256-D
Gated
fusion
110-speaker
classifier